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Food container and animal food dispensing device compatible with food container

Abstract

A food container and an animal food dispensing device compatible with food container are provided. The food container includes a can body and a cover body. The can body includes a container body including a container bowl portion. The container bowl portion is defined with an accommodation chamber and an opening communicated with the accommodation chamber. The cover body includes a side wall portion, a cover tray portion, and a cover flange portion. The cover tray portion allows the cover body, along an up-down direction of the cover body, to be defined with a recessed space having an opening opened upward and a lower space having an opening opened downward. The cover flange portion, the side wall portion, and the cover tray portion define a groove. The cover body is assembled to the can body, and the container body is snap-fitted with the side wall portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present disclosure is a Continuation Application of PCT Application No. PCT/CN2024/080294, filed Mar. 6, 2024, which claims the priority of Chinese Patent Application No. 202420273925.2, filed Feb. 4, 2024 with the National Intellectual Property Administration and entitled “food container and animal food dispensing device compatible with food container”, the entirety of which is hereby incorporated herein by reference for all purposes.

TECHNICAL FIELD

(1) The present disclosure relates to the technical field of food containers for pets, and in particular to a food container and an animal food dispensing device compatible with food container.

BACKGROUND

(2) Pets are becoming indispensable companions in people's lives. Nowadays, the number of pet

owners is on the rise. When feeding pets, in addition to cat food, pet canned food and other types of food are also provided. Food containers, such as pet cans, are generally used to hold wet food, such as minced meat, but of course, can also be used for holding dry food. For this type of food container, the cover is typically integrated with the entire container and is opened by means of a pull ring, which is inconvenient and is prone to accidents, and is not suitable for pet feeders that need to automatically open the containers.

SUMMARY

(3) The present disclosure aims to address the technical problems in related art. Therefore, the present disclosure provides a food container suitable for an automatic can-opening mechanism and with a cover that is easy to open.

(4) To realize the above objective, according to a first aspect, the present disclosure provides a food container, including: a can body, the can body including a container body, the container body comprising a container bowl portion, and the container bowl portion being defined with an accommodation chamber for holding food and an opening opened upward and communicated with the accommodation chamber; and a cover body, the cover body including a side wall portion, a cover tray portion, and a cover flange portion, the cover tray portion being connected to an inner peripheral wall of the side wall portion, the cover tray portion allowing the cover body, along an up-down direction of the cover body, to be defined with a recessed space having an opening opened upward and a lower space having an opening opened downward, the cover flange portion being arranged in the recessed space, an end of the cover flange portion being connected to the side wall portion, the other end of the cover flange portion extending inward in a direction away from the side wall portion, and the cover flange portion, the side wall portion and the cover tray portion defining a groove; the cover body is assembled to the can body, and the container body is snap-fitted with the side wall portion.

(5) In some embodiments, the food container further includes a sealing membrane for sealing the accommodation chamber. A portion of the sealing membrane is connected to the cover tray portion, and at least part of the other portion of the sealing membrane is connected to the container bowl portion.

(6) In some embodiments, the sealing membrane includes a sealing membrane body and a connection lug portion extending outward from an outer peripheral edge of the sealing membrane body. The sealing membrane body is connected to an upper surface of the container bowl portion, and the connection lug portion is folded upward and inward to be connected to a lower surface of the cover tray portion.

(7) In some embodiments, the connection lug portion includes a first adhesion section and a second adhesion section connected to the first adhesion section, and the first adhesion section is closer to an outer peripheral edge of the sealing membrane body than the second adhesion section. The connection lug portion is one of two connection lug portions, the first adhesion section of one of the two connection lug portions is connected to the lower surface of the cover tray portion, and the second adhesion section of the other one of the two connection lug portions is connected to the lower surface of the cover tray portion.

(8) In some embodiments, the sealing membrane has a first surface and a second surface facing the first surface, the first surface of the sealing membrane is connected to an upper surface of the container bowl portion, and the second surface of the sealing membrane is connected to a lower surface of the cover tray portion. Or, the cover tray portion is defined with a first through hole, the sealing membrane includes a sealing membrane body and a connection lug portion extending outward from an outer peripheral edge of the sealing membrane body, the sealing membrane body is connected to an upper surface of the container bowl portion, the connection lug portion passes through the first through hole into the recessed space, and the connection lug portion is connected to an upper surface of the cover tray portion.

(9) In some embodiments, the cover body is further provided with a handle portion arranged in the

recessed space. The handle portion is connected to the cover tray portion, and an upper surface of the handle portion is lower than the cover flange portion in an axial direction of a central axis of the cover body.

(10) In some embodiments, the cover tray portion is defined with a second through hole, and the handle portion is arranged above the second through hole.

(11) In some embodiments, the second through hole is located at a center of the cover body, within a projection of the food container on a horizontal plane.

(12) In some embodiments, the cover body is further provided with a first flange portion arranged in the lower space, an end of the first flange portion is connected to the side wall portion, and the other end of the first flange portion extends inward in a direction away from the side wall portion. The container body is further provided with a second flange portion, an end of the second flange portion is connected to the container bowl portion, and the other end of the second flange portion extends outward in a direction away from the container body. The cover body is assembled to the can body, and the first flange portion is snap-fitted with the second flange portion.

(13) In some embodiments, the second flange portion forms an upper peripheral edge of the container body. The cover body is assembled to the can body, and an inner edge of the cover flange portion fails to protrude beyond an inner surface of the second flange portion, within a projection of the food container on a horizontal plane.

(14) In some embodiments, the food container is configured to hold dry food or wet food. And/or, the cover flange portion forms an upper peripheral edge of the cover body. And/or, the groove is an annular groove. And/or, the food container is configured to be compatible with an automatic feeding assembly.

(15) In the technical solution of the present disclosure, the container bowl portion is provided in the can body for holding food, and the cover tray portion of the cover body covers the opening of the container bowl portion. When the cover body is assembled on the can body, the container body is snap-fitted with the side wall portion to facilitate the opening of the cover body. Moreover, the cover flange portion, the side wall portion, and the cover tray portion form the groove, which is convenient for an external animal food dispensing device to be connected to the cover body, to drive the cover body away from the can body. Therefore, the food container provided by the present disclosure can be used with the external animal food dispensing device to automatically open the cover body.

(16) According to a second aspect, the present disclosure provides an animal food dispensing device compatible with food container. The animal food dispensing device is applied to the food container according to any of the foregoing embodiments. The food container further includes a sealing membrane for sealing the accommodation chamber. a portion of the sealing membrane is connected to the cover body, and another portion of the sealing membrane is connected to the can body. The animal food dispensing device includes a press-can apparatus and a pull-cover apparatus. The press-can apparatus is configured to abut against the can body to limit movement of the can body in a vertical direction.

(17) The pull-cover apparatus includes: a pull-cover body, the pull-cover body being movably connected with a snap-fit assembly, the snap-fit assembly having a locked state and a released state, and the snap-fit assembly being switchable between the locked state and the released state; in the locked state, the snap-fit assembly being engaged in the groove, and in the released state, the snap-fit assembly being separated from the groove; and a first driving assembly, the first driving assembly being in transmission connection with the pull-cover body, to allow the pull-cover body to have a first position and a second position in the vertical direction; at the first position, the snap-fit assembly being configured to engage with the groove, and at the second position, the cover body being separated from the can body and driving the sealing membrane to separate from the can body; and the first driving assembly driving the pull-cover body to move between the first position and the second position.

(18) In some embodiments, the pull-cover apparatus further includes a second driving assembly. The second driving assembly includes a second motor and a first cam. The second motor is connected to the first cam to drive the first cam to rotate, an outer peripheral wall of the first cam is provided with a protruding portion extending in a direction perpendicular to an axial direction of the second motor and a recessed portion. The snap-fit assembly includes an engaging member. The engaging member includes an engaging portion and a pushing portion connected to each other. In the locked state, the first cam drives the protruding portion to abut against the pushing portion, and the pushing portion drives the engaging portion to engage with the groove; and in the released state, the first cam drives the recessed portion to abut against the pushing portion, and the pushing portion drives the engaging portion to separate from the groove.

(19) In some embodiments, the snap-fit assembly further includes a reset member. The engaging member is further provided with a fixing portion extending upward along an axial direction of a motor shaft of the second motor. An end of the reset member is connected with the pull-cover body, and the other end of the reset member is connected with the fixing portion. The reset member applies an elastic restoring force to the engaging member, to allow the engaging member to switch from the locked state to the released state. And/or, the pull-cover body is in a shape of a sleeve with an internal first accommodation space, the second driving assembly and the snap-fit assembly are arranged in the first accommodation space, a guide rail is provided on a lower wall surface of the first accommodation space, and the engaging member is in sliding fit with the guide rail. And/or, the engaging portion is provided with a first guide inclined surface and a support surface connected with the first guide inclined surface. The first guide inclined surface extends upward from an edge of the engaging member to the support surface; and the first guide inclined surface is configured to cooperate with the cover body during the pull-cover body moving to the first position, to guide the cover body into the support surface.

(20) In some embodiments, the second driving assembly further includes a second cam. The first cam includes a cam shaft. The second cam is sleeved on the cam shaft, and the second motor drives the first cam and the second cam to move synchronously. The animal food dispensing device further includes a first detection assembly. The first detection assembly includes a first signal emitter and a first signal receiver. The first signal emitter is fixedly connected to the second cam, the first signal receiver is fixedly connected to the pull-cover body, and in the locked state or/and the released state, the first signal emitter is provided in correspondence with the first signal receiver.

(21) In some embodiments, the first detection assembly is a mechanical switch, the first signal emitter is a mechanical protrusion, and the first signal receiver is a mechanical button. And/or, the first detection assembly includes two first signal receivers. The number of the first signal emitters is the same as the number of the engaging members and is denoted as N . The first signal emitters and the engaging members are evenly spaced along an axis of the second motor, within a projection on a horizontal plane. The line connecting the axis of the second motor to one of the first signal receivers is $L1$, and the line connecting the axis of the second motor to the other first signal receiver is $L2$, where the central angle defined between $L1$ and $L2$ is α , and α and N satisfy the equation: $\alpha=180/N$.

(22) In some embodiments, a bottom end of the pull-cover body is provided with a guide portion for cooperating with the cover body. The guide portion is configured to move in an inner peripheral wall of the cover body, to allow the pull-cover body to move into or out of the cover body. The guide portion is defined with a first through hole, and in the locked state, the engaging portion extends out of the first through hole to engage with the groove.

(23) In some embodiments, the guide portion is provided with a second guide inclined surface, the second guide inclined surface is inclined from a bottom of the pull-cover body in a vertical upward direction toward a direction close to the press-can apparatus, and the second guide inclined surface is configured to cooperate with the cover body during the pull-cover body moving to the first

position, to allow the pull-cover body to move into the cover body.

(24) In some embodiments, the press-can apparatus includes a pressing member and a third driving assembly. The third driving assembly is in transmission connection with the pressing member, to allow the pressing member to have a third position and a fourth position in the vertical direction. At the third position, the pressing member abuts against the first flange portion to limit movement of the can body in the vertical direction; and at the fourth position, the pressing member is separated from the first flange portion. The third driving assembly drives the pressing member to move between the third position and the fourth position.

(25) In some embodiments, the first driving assembly includes a first motor and a first transmission member in transmission connected with the first motor. The first transmission member is fixedly connected with the pull-cover body, and the first motor drives the first transmission member to move in the vertical direction. The pressing member is in a shape of a sleeve with an internal second accommodation space, an end of the pressing member close to the food container is open, and the other end of the pressing member is defined with a second through hole penetrating up and down, the pull-cover body is arranged in the second accommodation space and the first transmission member passes through the second through hole to drive the pull-cover body to move in the vertical direction.

(26) In some embodiments, the animal food dispensing device further includes a fixing member. The pressing member and the pull-cover body are fixedly connected with the fixing member. The fixing member is opened with a second through hole.

(27) In some embodiments, the first transmission member includes a first gear and a first rack. The first motor is mounted on the fixing member. An output shaft of the first motor is in transmission connection with the first gear. The first rack has a first surface and a second surface facing the first surface. The first surface is meshed with the first gear for transmission. The surface of the fixing member facing the first motor is provided with a limiting portion, and the limiting portion is arranged close to the second surface to limit the first rack to move in a direction away from the first gear. The first rack passes through the second through hole to drive the pull-cover body to move vertically.

(28) In some embodiments, the limiting portion includes a third roller, which is cooperated with the first rack to reduce friction.

(29) In some embodiments, the animal food dispensing device further includes a second detection assembly. The second detection assembly includes a second signal emitter and a second signal receiver. The second signal emitter is fixedly connected to the first rack, and the second signal receiver is fixedly connected to the fixing member. The second signal emitter is a mechanical protrusion, and the second signal receiver is a mechanical button. At the first position, the second signal emitter abuts against the second signal receiver. And/or, the bottom of the pressing member is provided with a flanging extending outward along the circumferential direction. The flanging is configured to abut against the support portion of the can body. And/or, the animal food dispensing device further includes a first walking assembly. The first walking assembly includes a first roller, which is assembled to the pull-cover body. The first roller is in rolling cooperation with the pressing member to reduce friction generated when the pull-cover body and the pressing member move relative to each other. And/or, the third position is below the fourth position.

(30) In some embodiments, the third driving assembly includes a third motor and a second transmission member. The second transmission member includes a second gear and a second rack. An output shaft of the third motor is in transmission connection with the second gear, the pressing member is fixedly connected with the second rack, and the second gear is meshed with the second rack to drive the pressing member to move in the vertical direction.

(31) In some embodiments, the animal food dispensing device further includes a housing in a shape of a sleeve with an internal third accommodation space. The third motor is mounted on the housing, and the press-can apparatus and the pull-cover apparatus are arranged in the third accommodation

space. The animal food dispensing device further includes a second walking assembly. The second walking assembly includes a second roller, which is assembled on the housing. The second roller is in rolling operation with the pressing member to reduce friction generated when the pressing member moves relative to the housing.

(32) In some embodiments, the animal food dispensing device further includes a third detection assembly. The third detection assembly includes a third signal emitter and two third signal receivers. The third signal emitter is fixedly connected to the press-can apparatus, and the third signal receivers are fixedly connected to the housing. At the third position, one of the two third signal receivers is provided in correspondence with the third signal receiver; and at the fourth position, the other of the two third signal receivers is provided in correspondence with the third signal receiver.

(33) In some embodiments, the third detection assembly is a photoelectric sensor, the third signal emitter is a photoelectric emitter, and the third signal receiver is a photoelectric receiver; or, the third detection assembly is a magnetic field sensor, the third signal emitter is a magnetic signal emitter, and the third signal receiver is a magnetic signal receiver; or, the third detection assembly is a mechanical switch, the third signal emitter is a mechanical protrusion, and the third signal receiver is a mechanical button.

(34) In the technical solution of the present disclosure, the container bowl portion is provided in the can body for holding food, and the cover tray portion of the cover body covers the opening of the container bowl portion. When the cover body is assembled on the can body, the container body is snap-fitted with the side wall portion to facilitate the opening of the cover body. Moreover, the cover flange portion, the side wall portion, and the cover tray portion form the groove, which is convenient for an external animal food dispensing device to be connected to the cover body, to drive the cover body away from the can body. Therefore, the food container provided by the present disclosure can be used with the external animal food dispensing device to automatically open the cover body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to explain the technical solutions of the embodiments of the present disclosure more clearly, the following accompanying drawings are briefly described. Obviously, the accompanying drawings are only some embodiments of the present disclosure, for those skilled in the field, other drawings can be obtained based on these drawings without any creative effort.

(2) FIG. 1 is a schematic structural diagram of a food container according to an embodiment of the present disclosure.

(3) FIG. 2 is an exploded structural diagram of a food container according to an embodiment of the present disclosure.

(4) FIG. 3 is a cross-sectional diagram of a food container according to an embodiment of the present disclosure.

(5) FIG. 4 is a schematic structural diagram of a cover body of a food container according to an embodiment of the present disclosure.

(6) FIG. 5 is an exploded structural diagram of a cover body of a food container according to an embodiment of the present disclosure.

(7) FIG. 6 is a cross-sectional diagram of a cover body of a food container according to an embodiment of the present disclosure.

(8) FIG. 7 is a cross-sectional diagram illustrating stacking of multiple cover bodies of a food container according to an embodiment of the present disclosure.

(9) FIG. 8 is an exploded structural diagram of a can body of a food container according to an

embodiment of the present disclosure.

(10) FIG. 9 is a cross-sectional diagram of a can body of a food container according to an embodiment of the present disclosure.

(11) FIG. 10 is a top view of a can body of a food container according to an embodiment of the present disclosure.

(12) FIG. 11 is a schematic structural diagram of a food container according to another embodiment of the present disclosure.

(13) FIG. 12 is a cross-sectional diagram of a food container according to another embodiment of the present disclosure.

(14) FIG. 13 is an enlarged view of portion A in FIG. 12.

(15) FIG. 14 is a schematic structural diagram of a sealing membrane and a can body of a food container according to another embodiment of the present disclosure.

(16) FIG. 15 is a cross-sectional diagram of an animal food dispensing device compatible with food container at a first position according to an embodiment of the present disclosure.

(17) FIG. 16 is a cross-sectional diagram of an animal food dispensing device compatible with food container at a second position according to an embodiment of the present disclosure.

(18) FIG. 17 is schematic structural diagram of a snap-fit assembly of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure.

(19) FIG. 18 is a schematic structural diagram of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure.

(20) FIG. 19 is an exploded structural diagram of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure.

(21) FIG. 20 is a cross-sectional diagram of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure.

(22) FIG. 21 is a cross-sectional diagram of an animal food dispensing device compatible with food container according to another embodiment of the present disclosure.

(23) FIG. 22 is a schematic structural diagram of a pull-cover body of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure from a perspective.

(24) FIG. 23 is a schematic structural diagram of a pull-cover body of an animal food dispensing device compatible with food container according to an embodiment of the present disclosure from another perspective.

(25) The purpose, functional features, and advantages of the present disclosure will be further illustrated in conjunction with embodiments, with reference to the drawings.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(26) The technical solutions in the embodiments of the present disclosure will be described clearly and completely with reference to the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only a part of the embodiments of the present disclosure rather than all of them. Based on the embodiments in the present disclosure, all other embodiments obtained by those skilled in the art without creative work shall fall within the scope of protection of the present disclosure.

(27) It should be noted that all directional indications (such as up, down, left, right, front, back) in the embodiments of the present disclosure are merely configured to explain relative position relationships or motion conditions between the components in a specific attitude (as shown in the drawings). The directional indication changes as the specific attitude changes.

(28) It should be noted that unless otherwise explicitly defined and limited, terms such as “connected” and “fixed” should be understood in a broad sense. For example, “fixed” can mean a rigid connection, a detachable connection, or being integrated into one piece; it may be a mechanical connection or an electrical connection; it may be a direct connection or an indirect connection through an intermediary; it may be an internal connection within two components or an

interactive relationship between two components, unless otherwise specifically limited. For those skilled in the art, the specific meaning of the aforementioned terms in the present disclosure can be understood according to the specific circumstances.

(29) Moreover, the terms “first”, “second”, and the like in the present disclosure are merely used for description and cannot be understood as indicating or implying their relative importance or as implicitly indicating the quantity of the technical features indicated. Thus, the feature defined by “first” or “second” may explicitly or implicitly include at least one such feature. Furthermore, the term “and/or” throughout the description includes three concurrent scenarios. For example, “A and/or B” includes scenario A alone, scenario B alone, or both A and B together. Additionally, the technical solutions of various embodiments may be combined with each other, but must be based on that the combined technical solutions can be implemented by those skilled in the art. When the combination of the technical solutions is contradictory or impossible to realize, it shall be considered that such combination does not exist and is not within the scope of protection of the present disclosure.

(30) The present disclosure provides a food container. In related art, the cover of the food container is inconvenient to open and is not suitable for an animal food dispensing device that automatically opens the food container.

(31) To solve the above technical problems, in the technical solution of the present disclosure, the container bowl portion is provided in the can body for holding food, and the cover tray portion of the cover body covers the opening of the container bowl portion. When the cover body is assembled on the can body, the container body is snap-fitted with the side wall portion to facilitate the opening of the cover body. Moreover, the cover flange portion, the side wall portion, and the cover tray portion form the groove, which is convenient for an external animal food dispensing device to be connected to the cover body, to drive the cover body away from the can body. Therefore, the food container provided by the present disclosure can be used with the external animal food dispensing device to automatically open the cover body.

(32) The above technical solutions will be described in detail below with reference to the accompanying drawings.

(33) In the embodiments, as shown in FIG. 1 to FIG. 6, the food container includes a can body **1000** and a cover body **2000**. The can body **1000** includes a container body **1100**. The container body **1100** includes a container bowl portion **1110**. The container bowl portion **1110** is defined with an accommodation chamber for holding food and an opening opened upward and communicated with the accommodation chamber. It should be noted that food is placed in the container bowl portion **1110**, and the cover body **2000** is fastened on the can body **1000**. The can body **1000** is detachably connected with the cover body **2000**. When it is necessary to feed a pet with food from the container bowl portion **1110**, the cover body **2000** is driven to separate from the can body **1000**, exposing the opening. In this way, the pet can consume the food placed in the container bowl portion **1110** through the opening. In a specific implementation process, the accommodation chamber of the container bowl portion **1110** can be configured to hold dry food or wet food.

(34) Referring to FIG. 4 to FIG. 6, the cover body **2000** includes a side wall portion **2100**, a cover tray portion **2200**, and a cover flange portion **2300**. The cover tray portion **2200** is connected to an inner peripheral wall of the side wall portion **2100**. The cover tray portion **2200** allows the cover body **2000**, along an up-down direction of the cover body **2000**, to be defined with a recessed space having an opening opened upward and a lower space having an opening opened downward. The cover flange portion **2300** is arranged in the recessed space. One end of the cover flange portion **2300** is connected to the side wall portion **2100**, and the other end of the cover flange portion **2300** extends inward in a direction away from the side wall portion **2100**. A groove **2400** is defined by the cover flange portion **2300**, the side wall portion **2100**, and the cover tray portion **2200**. When the cover body **2000** is assembled to the can body **1000**, the container body **1100** is snap-fitted with the side wall portion **2100**.

(35) In a specific implementation process, the shape of the cover tray portion **2200** is adapted to fit the opening of the accommodation chamber. The side wall portion **2100** is arranged along the side wall edge of the cover tray portion **2200** and surrounds the cover tray portion **2200**. The upper and lower ends of the side wall portion **2100** are respectively located on two sides of the cover tray portion **2200**, to form the recessed space on one side of the cover tray portion **2200** and the lower space on the other side. An opening is defined in the upper part of the recessed space, which is convenient for the animal food dispensing device to connect with the cover body **2000** and drive the cover body **2000** to separate from the can body **1000**. An opening is defined in the lower part of the lower space, and the top of the can body **1000** is accommodated in the lower space through the opening. The edge of the cover flange portion **2300** is connected to the top edge of the side wall portion **2100** and extends in a direction close to the axis of the cover body **2000**, such that the groove **2400** is formed by the cover flange portion **2300**, the side wall portion **2100** located on the side of the recessed space, and the cover tray portion **2200**. The cover flange portion **2300** forms the upper peripheral edge of the cover body **2000**. The animal food dispensing device can extend into the groove **2400** and is engaged with the cover body **2000** to pull the cover body **2000**. This allows the cover body **2000** to be separated from the can body **1000**, thereby opening the food container. It should be noted that, in order for the animal food dispensing device to effectively engage with the cover body **2000**, in a specific implementation process, the cover flange portion **2300** extends in the direction close to the axis of the cover body **2000**, and the cover flange portion **2300** is horizontally arranged. Specifically, the bottom wall of the cover flange portion **2300** facing the groove **2400** is horizontally arranged, enhancing the stability of the engagement.

(36) The side wall portion **2100** is configured to be snap-fitted with the container body **1100**, thereby realizing the detachable connection between the cover body **2000** and the can body **1000**. The snap-fit between the side wall portion **2100** and the container body **1100** may be implemented by a snap-fit structure. For example, one of the side wall portion **2100** and the container body **1100** is provided with a protrusion, and the other one is defined with a groove snap-fitted with the protrusion; or, one of the side wall portion **2100** and the container body **1100** is connected with a snap fastener, and the other one is connected with another snap fastener mated with the aforementioned snap fastener, which is not limited herein.

(37) As shown in FIG. 3 and FIG. 6, the cover body **2000** is further provided with a first flange portion **2600**. The first flange portion **2600** is arranged in the lower space. One end of the first flange portion **2600** is connected to the side wall portion **2100**, and the other end of the first flange portion **2600** extends inward in a direction away from the side wall portion **2100**. The container body **1100** is further provided with a second flange portion **1120**. One end of the second flange portion **1120** is connected to the container bowl portion **1110**, and the other end of the second flange portion **1120** extends outward in a direction away from the container body **1100**. When the cover body **2000** is assembled to the can body **1000**, the first flange portion **2600** is snap-fitted with the second flange portion **1120**.

(38) In the embodiments, the cover body **2000** is snap-fitted with the can body **1000** by the first flange portion **2600** and the second flange portion **1120**. In some embodiments, the first flange portion **2600** is a protrusion that protrudes from the side wall portion **2100** and is located on the side of the side wall portion **2100** facing the lower space. The second flange portion **1120** is a protrusion protruding from the container bowl portion **1110** or a protruding strip structure surrounding the peripheral side wall. The first flange portion **2600** is snap-fitted with the second flange portion **1120**. In another embodiment, one end of the first flange portion **2600** is a fixed end, and the other end of the first flange portion **2600** is a free end. The fixed end is connected to the side wall portion **2100**, and the free end extends away from the surface of the side wall portion **2100**, such that the first flange portion **2600** protrudes from the surface of the side wall portion **2100**. The second flange portion **1120** is arranged with reference to the first flange portion **2600**, to ensure smooth snap-fit between the first flange portion **2600** and the second flange portion **1120**.

(39) In a specific implementation process, to enhance the strength and effectiveness of the snap-fit, the first flange portion **2600** is integrally formed with the side wall portion **2100**, and the second flange portion **1120** is integrally formed with the container bowl portion **1110**. Moreover, to enhance the strength of the snap-fit, multiple first flange portions **2600** are arranged at intervals along the circumferential direction of the cover body **2000**, and the second flange portion **1120** extends along the axis of the container bowl portion **1110** and surrounds the container bowl portion **1110**.

(40) Further, based on the aforementioned, in the embodiments, the second flange portion **1120** forms the upper peripheral edge of the container body **1100**. When the cover body **2000** is assembled to the can body **1000**, the inner edge of the cover flange portion **2300** does not protrude beyond the inner surface of the second flange portion **1120**, within a projection of the food container on a horizontal plane. According to the embodiments, when the width of the engagement part of the groove **2400** is constant, compared to other solutions (e.g., the inner edge of the cover flange portion **2300** protrudes beyond the inner surface of the second flange portion **1120**), the stacking of multiple cover bodies **10** is more stable, thus being beneficial for transportation and assembly in the factory. The stacking of multiple cover bodies **10** is shown in FIG. 7.

(41) It can be understood that bacteria and moisture in the air can affect the food in the container bowl portion **1110**, making it prone to spoilage. Therefore, it is necessary to seal the containment chamber of the container bowl portion **1110** to extend the shelf life of the food. In some embodiments, as shown in FIG. 5 and FIG. 6, the food container further includes a sealing membrane **3000** for sealing the accommodation chamber, to improve the sealing effect of the accommodation chamber. In addition, when opening the food container, it is necessary to pull the cover body **2000** and the sealing membrane **3000** in sequence, which is inconvenient, especially not suitable for the animal food dispensing device that automatically opens the food container. Further, in the embodiments, a portion of the sealing membrane **3000** is connected to the cover tray portion **2200**, and at least part of the other portion of the sealing membrane **3000** is connected to the container bowl portion **1110**. In this way, the sealing membrane **3000** is connected to the cover tray portion **2200** while sealing the accommodation chamber. Thus, when the cover body **2000** is driven to separate from the can body **1000**, the sealing membrane **3000** is also separated from the container bowl portion **1110** to expose the food, eliminating the need to open the sealing membrane **3000** again. Moreover, the sealing membrane **3000** may be connected by means of adhesive attachment or heat fusion, which depends on an actual requirement and the material of the sealing membrane **3000**. In a specific implementation process, to ensure that the sealing membrane **3000** has a certain strength to avoid damage, an aluminum membrane is used for the sealing membrane **3000**.

(42) Specifically, in some embodiments, the sealing membrane **3000** has a first surface and a second surface facing the first surface. The first surface of the sealing membrane **3000** is connected to an upper surface of the container bowl portion **1110**, and the second surface of the sealing membrane **3000** is connected to a lower surface of the cover tray portion **2200**. In some embodiments, the first surface of the sealing membrane **3000** is connected to the upper surface of the container bowl portion **1110** by means of adhesive attachment. It can be understood that, a ring of adhesive is applied on the part of the first surface of the sealing membrane **3000** close to its edge along the perimeter, and it is bonded to the upper surface of the container bowl portion **1110**. An adhesive may be applied at any position on the second surface to bond with the lower surface of the cover tray portion **2200**. The strength of the adhesive bonding between the second surface and the cover tray portion **2200** ensures that when the cover body **2000** is driven to move away, the sealing membrane **3000** can release its adhesion to the container bowl portion **1110** and move synchronously with the cover body **2000**.

(43) In another embodiment, referring to FIG. 5, the cover tray portion **2200** is defined with a first through hole **2210**, and the sealing membrane **3000** includes a sealing membrane body **3100** and a

connection lug portion **3200** extending outward from an outer peripheral edge of the sealing membrane body **3100**. The sealing membrane body **3100** is connected to the upper surface of the container bowl portion **1110**. The connection lug portion **3200** passes through the first through hole **2210** into the recessed space, and the connection lug portion **3200** is connected with the upper surface of the cover tray portion **2200**.

(44) In a specific implementation process, to improve the strength of the sealing membrane **3000**, the connection lug portion **3200** is integrally formed with the sealing membrane body **3100**. It can be understood that the part of the sealing membrane body **3100** close to its edge is connected with the top surface of the container bowl portion **1110**. To allow the cover body **2000** to smoothly drive the sealing membrane **3000** to move, the connection lug portion **3200** is connected with the edge of the sealing membrane body **3100**, to allow the force of the cover body **2000** to directly act on the edge of the sealing membrane body **3100**, further releasing the connection between the sealing membrane body **3100** and the container bowl portion **1110**. The cover tray portion **2200** is provided with the first through hole **2210** close to its edge, the end of the connection lug portion **3200** away from the sealing membrane body **3100** enters the recessed space through the first through hole **2210**, and one surface of the connection lug portion **3200** is connected to the surface of the cover tray portion **2200** facing the recessed space. By using the connection lug portion **3200**, the pulling force of the cover body **2000** can be directly applied to the connection of the sealing membrane body **3100** and the container bowl portion **1110**, making it easier to release the connection between the two. Further, to ensure balanced and effective separation of the sealing membrane body **3100** from the container bowl portion **1110**, in a specific implementation process, the sealing membrane body **3100** is provided with at least two sets of connection lug portions **3200**, which are symmetrically arranged.

(45) Referring to FIG. **11** to FIG. **14**, in another embodiment of the food container, the sealing membrane **3000** includes a sealing membrane body **3100** and a connection lug portion **3200** extending outward from an outer peripheral edge of the sealing membrane body **3100**. The sealing membrane body **3100** is connected with the upper surface of the container bowl portion **1110**, and the connection lug portion **3200** is folded upward and inward to connect with the lower surface of the cover tray portion **2200**. It can be understood that the part of the sealing membrane body **3100** close to its edge is connected with the top surface of the container bowl portion **1110**. To allow the cover body **2000** to smoothly drive the sealing membrane **3000** to move, the connection lug portion **3200** is connected with the edge of the sealing membrane body **3100**, and the connection lug portion **3200** is folded upward and inward to connect with the cover body **2000**. This allows the force of the cover body **2000** to be transferred to the edge of the sealing membrane body **3100** via the connection lug portion **3200**, thereby further releasing the connection between the sealing membrane body **3100** and the container bowl portion **1110**.

(46) In the embodiments, as shown in FIG. **14**, the connection lug portion **3200** includes a first adhesion section **3210** and a second adhesion section **3220** connected to each other. The first adhesion section **3210** is closer to the outer peripheral edge of the sealing membrane body **3100** than the second adhesion section **3220**. The sealing membrane **3000** includes two connection lug portions **3200**, the first adhesion section **3210** of one of the two connection lug portions **3200** is connected to the lower surface of the cover tray portion **2200**, and the second adhesion section **3220** of the other one of the two connection lug portions **3200** is connected to the lower surface of the cover tray portion **2200**. It will be appreciated that the upward movement of the cover body **2000** causes the two connection lug portions **3200** of the sealing membrane **3000** to be subjected to different forces, which is advantageous for maintaining the structural stability of the food container.

(47) To ensure effective separation of the sealing membrane body **3100** from the container bowl portion **1110**, in a specific implementation process, the sealing membrane **3000** includes multiple connection lug portions **3200**, which are symmetrically arranged around the central axis of the sealing membrane **3000**.

(48) As shown in FIG. 13, when the sealing membrane 3000 is assembled to the can body 1000 and the cover body 2000, the outer edge of the sealing membrane 3000 (such as the part where the connection lug portion 3200 is folded when the second adhesion section 3220 of the connection lug portion 3200 is connected with the lower surface of the cover tray portion 2200) protrudes beyond the outer peripheral surface of the container body 1100, within a projection of the food container on a horizontal plane. The can body 1000 covers and accommodates the outer part of the sealing membrane 3000 protruding beyond the outer peripheral surface of the container body 1100.

(49) Certainly, based on the foregoing embodiments, it can be understood that the connection between the sealing membrane 3000 and the container bowl portion 1110, as well as between the sealing membrane 3000 and the cover tray portion 2200 may be realized by means of vacuum suction, as long as the accommodation chamber of the container bowl portion 1110 maintains a certain level of airtightness.

(50) In some embodiments, the cover body 2000 is further provided with a handle portion 2500. The handle portion 2500 is arranged in the recessed space, and is connected to the cover tray portion 2200. The upper surface of the handle portion 2500 is lower than the cover flange portion 2300 in the axial direction of the central axis of the cover body 2000.

(51) The handle portion 2500 is configured for hand-holding, which is convenient for lifting the cover body 2000. In order to improve the strength of the handle portion 2500, in a specific implementation process, the handle portion 2500 is integrally formed with the cover tray portion 2200. In practice, the handle portion 2500 is an arc-shaped strip structure, with two ends connected to the cover tray portion 2200. In addition, an extension direction from either end toward the center is an arc, allowing the middle part of the handle portion 2500 to protrude upward and be higher than the two ends. This creates a certain space between the handle portion 2500 and the cover tray portion 2200, to facilitate the user's hand to reach into the space between the handle portion 2500 and the cover tray portion 2200, thereby lifting the handle portion 2500.

(52) Further, the cover tray portion 2200 is defined with a second through hole 2220, and the handle portion 2500 is arranged above the second through hole 2220. Specifically, the handle portion 2500 is arranged across the second through hole 2220 to increase the space below the handle portion 2500, facilitating the user's hand to pull the cover body 2000.

(53) In a specific implementation process, within a projection of the food container on a horizontal plane, the second through hole 2220 is located at the center of the cover body 2000, and the handle portion 2500 is arranged across the second through hole 2220, with two ends being at an equal distance from the second through hole 2220. In this way, the center of the handle portion 2500 is located at the center of the cover body 2000. By using the handle portion 2500 to pull the cover body 2000, the force can be evenly transferred to the edge of the cover body 2000, which allows the cover body 2000 to smoothly separate from the can body 1000, enhancing the stability of opening the food container.

(54) Referring to FIG. 8 to FIG. 10, a support member 1200 is fixedly sleeved on the outer peripheral wall of the container bowl portion 1110, and one end of a third flange portion 1210 is connected with the container bowl portion 1110. The support member 1200 may be connected with the container bowl portion 1110 by means of adhesive fixation; or, the support member 1200 may be locked and fixed on the outer peripheral wall of the container bowl portion 1110 by a connection member such as a screw or a pin. In the embodiments, the support member is bonded to the container body; that is, adhesive dots are applied between the two to achieve a tight and secure bonding connection. In a specific implementation process, the support member 1200 is configured to abut against the animal food dispensing device for opening the food container. Specifically, the other end of the support member 1200 protrudes beyond the outer peripheral wall of the container body 1100, such that the part between two ends of the support member 1200 protrudes beyond the outer peripheral wall of the container body 1100, which facilitates the animal food dispensing device to abut against the support member 1200. In a specific implementation process, the outer

edge of the third flange portion **1210** protrudes beyond the outer peripheral surface of the cover body **2000** within a projection of the food container on a horizontal plane, preventing interference of the container bowl portion **1110**. This allows the animal food dispensing device, when pulling to open the cover body **2000**, to limit the movement of the can body **1000** by abutting against the support member **1200**, facilitating the animal food dispensing device to drive the cover body **2000** away from the can body **1000** so as to open the food container for pet feeding.

(55) In some embodiments, the support member **1200** is not integrally formed with the container body **1100**. It will be appreciated that the support member **1200** is fixedly connected to the container body **1100**, such that the animal food dispensing device abuts against the support member **1200** to limit the movement of the container body **1100**. In a specific implementation process, on the one hand, the production of the container body **1100** is independent from the production of the support member **1200**, the container body **1100** has its own mold, and the support member **1200** also has its own mold, which simplifies the design and manufacturing of the molds. On the other hand, the container bowl portion **1110** of the container body **1100** is used for holding food, which means that the material selection for the container body **1100** is quite strict to meet the hygiene requirements for pet food; while the support member **1200** does not come into contact with the food, so there is a wider variety of material options available for it. This allows for more implementation flexibility in the industrialization process.

(56) The cover body **2000** is snap-fitted with the container body **1100**. In some embodiments, the container body **1100** is further provided with the second flange portion **1120**. One end of the second flange portion **1120** is connected to the container bowl portion **1110**, and the other end of the second flange portion **1120** extends outward in a direction away from the container body **1100**. The outer edge of the second flange portion **1120** protrudes beyond the outer peripheral surface of the container bowl portion **1110** within a projection of the food container on a horizontal plane. The cover body **2000** is further provided with the first flange portion **2600**. One end of the first flange portion **2600** is connected to the side wall portion **2100**, and the other end of the first flange portion **2600** extends inward in a direction away from the side wall portion **2100**. When the cover body **2000** is assembled to the can body **1000**, the first flange portion **2600** is snap-fitted with the second flange portion **1120**.

(57) In a specific implementation process, the second flange portion **1120** is arranged around the peripheral side wall of the container body **1100**. In the embodiments, the second flange portion **1120** conforms to the contour of the outer peripheral wall of the container body **1100**. Specifically, the second flange portion **1120** is a ring-shaped structure, which is arranged around the edge of the container body **1100** close to the opening. The inner edge of the second flange portion **1120**, which is the ring-shaped structure, is connected with the outer peripheral side wall of the container bowl portion **1110**, and the outer edge of the second flange portion **1120** protrudes beyond the outer peripheral surface of the container bowl portion **1110**, such that the second flange portion extends horizontally away from the container bowl portion **1110**. It can be understood that the second flange portion **1120** can serve as a flanging structure on the upper end of the container bowl portion **1110**. Further, the second flange portion **1120** forms the upper peripheral edge of the container body **1100**. In practice, the second flange portion **1120** is integrally formed with the container bowl portion **1110**, thereby enhancing the overall strength and improving the connection strength between the cover body **2000** and the can body **1000**.

(58) In another embodiment, the support member **1200** is further provided with a fourth flange portion **1220**. The fourth flange portion **1220** forms the upper peripheral edge of the support member **1200**. The outer edge of the fourth flange portion **1220** protrudes beyond the outer peripheral surface of the container bowl portion **1110** within a projection of the food container on a horizontal plane. The cover body **2000** is further provided with a first flange portion **2600**. One end of the first flange portion **2600** is connected to the side wall portion **2100**, and the other end of the first flange portion **2600** extends inward in a direction away from the side wall portion **2100**. When

the cover body **2000** is assembled to the can body **1000**, the fourth flange portion **1220** is snap-fitted with the first flange portion **2600**.

(59) Referring to FIG. **8**, the fourth flange portion **1220** and the third flange portion **1210** are respectively located at the upper and lower ends of the support member **1200**. The fourth flange portion **1220** forms the upper peripheral edge of the support member **1200**, and the third flange portion **1210** forms the lower peripheral edge of the support member **1200**. The inner end of the fourth flange portion **1220** is close to the container bowl portion **1110**, and the other end extends away from the container bowl portion **1110**. This allows the outer edge of the fourth flange portion **1220** to protrude beyond the outer peripheral surface of the container bowl portion **1110**, which facilitates the snap-fit between the first flange portion **2600** and the fourth flange portion **1220**, and prevents interference from the container bowl portion **1110**.

(60) In the embodiments, the container body **1100** is further provided with a second flange portion **1120**. One end of the second flange portion **1120** is connected to the container bowl portion **1110**, and the other end of the second flange portion **1120** extends outward in a direction away from the container body **1100**. The upper surface of the fourth flange portion **1220** is connected to the lower surface of the second flange portion **1120**. The outer edge of the fourth flange portion **1220** protrudes beyond the outer peripheral surface of the second flange portion **1120** within a projection of the food container on a horizontal plane.

(61) Specifically, it should be noted that one end of the second flange portion **1120** is the side wall of the second flange portion **1120** close to the accommodation chamber, which is connected to the outer peripheral surface of the container bowl portion **1110** and extends circumferentially around the container bowl portion **1110**. The other end of the second flange portion **1120** extends in a direction away from the container bowl portion **1110** such that the other end of the second flange portion **1120** extends out of the container bowl portion **1110**. In practice, it can be understood that the second flange portion **1120** can serve as a flanging structure on the upper end of the container bowl portion **1110**, and the second flange portion **1120** is integrally formed with the container bowl portion **1110**, thereby enhancing the overall strength and improving the connection strength between the cover body **2000** and the can body **1000**. Further, the outer edge of the fourth flange portion **1220** protrudes beyond the outer peripheral surface of the second flange portion **1120**, facilitating the snap-fit with the cover body **2000** and providing protection to the second flange portion **1120**.

(62) Referring to FIG. **9**, in a specific implementation process, a second section **1112** extends in a horizontal direction, and the end portion of the second section **1112** close to the central axis of the container bowl portion **1110** forms a second peripheral wall. A first peripheral wall faces the second peripheral wall. Specifically, one end of the first section **1111** is connected with the first peripheral wall and extends upward to the opening of the accommodation chamber. One end of a third section **1113** is connected with the second peripheral wall and extends downward to the bottom wall of the container bowl portion **1110**. In this way, the first section **1111**, the second section **1112**, and the third section **1113** form the side wall of the container bowl portion **1110**, and a step structure is formed at the second section **1112**. This structure cooperates with a step portion **1230** of the support member, to allow the support member, namely the step portion **1230** and the parts extending upward and downward from the step portion **1230**, to connect with the side wall of the container bowl portion **1110**, thereby enhancing the connection strength between the support member **1200** and the container bowl portion **1110**.

(63) In practice, the food container is placed inside an animal food dispensing device to allow its cover body **2000** to be automatically opened. It can be understood that the animal food dispensing device is defined with a space or a recess for placing the food container. The animal food dispensing device needs to cooperate with the cover body **2000** and the can body **1000**, to allow the cover body **2000** to separate from the can body **1000**. There is a certain positioning requirement for the placement of the food container to ensure it is placed in the corresponding position each time

for proper cooperation with the animal food dispensing device. In the embodiments, the positioning requirement of the food container is achieved by providing a positioning portion **1240** on the support member **1200**.

(64) Specifically, in the embodiments, referring to FIG. **8**, the support member **1200** is further provided with a positioning portion **1240** extending outward in a direction away from the container body **1100**. The positioning portion **1240** is connected to an outer side of part of the third flange portion **1210**. Define points A and B as where the positioning portion **1240** is connected with the third flange portion **1210**, and point C as the outer edge point of the positioning portion **1240**. Then, the angle ACB is an acute angle, the length of the segment AC is greater than the length of the segment BC, and the segment AC is an arc segment. In a specific implementation process, to enhance the positioning effect, the support member **1200** is provided with two sets of positioning portions **1240**, which are arranged in central symmetry about the center of the support member **1200**. Specifically, there is a connecting line L between the midpoint of the point A and the point B and the central point of the container body **1100**. The point C of one set of positioning portion **1240** and the point C of the other set of positioning portion **1240** are located on two sides of the connecting line L. This arrangement helps prevent the food container from being inverted and ensures that the food container can be quickly positioned and accurately placed.

(65) Referring to FIG. **15** to FIG. **23**, the present disclosure further provides an animal food dispensing device compatible with food container. The animal food dispensing device is applied to the food container described above. The animal food dispensing device includes a press-can apparatus and a pull-cover apparatus. The press-can apparatus is configured to abut against the can body **1000**, to limit vertical movement of the can body **1000**. The press-can apparatus is located above the food container and abuts against the edge of the can body **1000**. Specifically, the press-can apparatus may be a sleeve structure, which uniformly abuts against the top surface of the outer side wall of the can body **1000**, thereby limiting the upward movement of the can body **1000**.

(66) The pull-cover apparatus includes a pull-cover body **100** and a first driving assembly **300**. The pull-cover body **100** is snap-fitted in the groove **2400** of the cover body **2000**. The first driving assembly **300** drives the pull-cover body **100** to move in the vertical direction. It should be noted that, in the embodiments, the cover body **2000** of the food container is snap-fitted with the can body **1000**, and through the snap-fit between the pull-cover body **100** and the groove **2400** of the cover body **2000**, the animal food dispensing device pulls the cover body **2000** away from the can body **1000** to open the food container. Specifically, the first driving assembly **300** drives the pull-cover body **100** to vertically pull the cover body **2000**, giving the cover body **2000** a tendency to move away from the can body **1000** in the vertical direction. Meanwhile, the press-can apparatus abuts against the can body **1000** to limit a vertical displacement of the can body **1000**. In this way, the pull-cover apparatus can pull the cover body **2000** away from the can body **1000** to open the food container.

(67) In a specific implementation process, with reference to FIG. **13**, the pull-cover body **100** is movably connected to a snap-fit assembly **200**. The snap-fit assembly **200** has a locked state and a released state, and it is switchable between the locked state and the released state. In the locked state, the snap-fit assembly **200** is engaged in the groove **2400**; and in the released state, the snap-fit assembly **200** is separated from the groove **2400**. Specifically, the snap-fit assembly **200** includes an engaging member **210**, a second driving assembly **400**, and a reset member **220**. The second driving assembly **400** is drivingly connected with the engaging member **210**, and drives one end of the engaging member **210** to extend out of the pull-cover body **100** to be engaged in the groove **2400** of the cover body **2000**. The reset member **220** is connected with the engaging member **210** and drives the engaging member **210** to move in the opposite direction, allowing the engaging member **210** to separate from the cover body **2000**. The second driving assembly **400** drives one end of the engaging member **210** to extend out of the pull-cover body **100**, such that the engaging member **210** is engaged with the cover body **2000**; and the first driving assembly **300**

cooperates with the reset member **220**, to allow the engaging member **210** to move away from the cover body **2000**, such that the engaging member **210** is separated from the cover body **2000**. This allows the snap-fit assembly **200** to switch between the locked state and the released state, thereby smoothly opening the food container. It can be understood that the reset member **220** has a certain elastic force. When the second driving assembly **400** drives the engaging member **210** to extend out, the reset member **220** is compressed, which allows it to have a reverse driving force.

Therefore, when the engaging member **210** needs to separate from the cover body **2000**, the reset member **220** can utilize the reverse driving force to drive the engaging member **210** to move reversely. In practice, the reset member **220** may be a spring, rubber, silicone, or the like, which is not limited herein. One end of the reset member **220** is connected to the engaging member **210**, and the other end of the reset member **220** is connected to the pull-cover body **100**.

(68) The transmission connection between the first driving assembly **300** and the pull-cover body **100** allows the pull-cover body **100** to have a first position and a second position in the vertical direction. At the first position, the snap-fit assembly **200** is configured to be engaged in the groove **2400**; and at the second position, the cover body **2000** is separated from the can body **1000** and drives the sealing membrane **3000** to separate from the can body **1000**. The first driving assembly **300** drives the pull-cover body **100** to move between the first position and the second position. Specifically, when it is necessary to open the food container, the first driving assembly **300** drives the pull-cover body **100** to reach the first position, and the bottom of the pull-cover body **100** enters the recessed space. The snap-fit assembly **200** is switched from the released state to the locked state and extends into the groove **2400** of the cover body **2000**, achieving the snap-fit. The first driving assembly **300** drives the pull-cover body **100** to move toward the second position such that the cover body **2000** moves away from the can body **1000** to separate from the can body **1000**.

Meanwhile, the sealing membrane **3000** is connected with the cover body **2000**, so as the cover body **2000** moves, it also drives the sealing membrane **3000** to separate from the can body **1000**, thereby opening the food container. In this way, the food in the container bowl portion **1110** is exposed for pets to eat. It can be understood that, the first driving assembly **300** may also drive the pull-cover body **100** to descend and carry the cover body **2000** to cover the can body **1000**, to protect the unfinished food inside the can body **1000** or facilitate the disposal of the entire food container. Specifically, in the process of the first driving assembly **300** driving the pull-cover body **100** and the cover body **2000** to descend to the first position, the snap-fit assembly **200** remains in the locked state to prevent the cover body **2000** from falling off. After reaching the first position, the snap-fit assembly **200** is switched from the locked state to the released state, which facilitates the pull-cover body **100** to separate from the cover body **2000**.

(69) In some embodiments, as shown in FIG. 15 to FIG. 23, the animal food dispensing device compatible with food container is applied to a can. The can includes a cover body **2000** and a can body **1000** snap-fitted with the cover body **2000**. The animal food dispensing device includes a press-can apparatus and a pull-cover apparatus. The press-can apparatus is configured to abut against the can body **1000** to limit vertical movement of the can body **1000**. The technical solution is applicable to the can whose cover body **2000** and can body **1000** are snap-fitted with each other. It can be understood that the cover body **2000** and the can body **1000** may move away from each other to open the can; or, one of the two is fixed while the other one moves away to open the can. In the embodiments, the press-can apparatus is configured to abut against the can body **1000** to limit the vertical displacement of the can body **1000**, and the pull-cover apparatus is configured to cooperate with the cover body **2000**, to allow the cover body **2000** to move vertically and separate from the can body **1000**, thereby opening the can.

(70) In the embodiments, the pull-cover apparatus includes a pull-cover body **100** and a first driving assembly **300**. Specifically, the pull-cover body **100** is movably connected with a snap-fit assembly **200**. The snap-fit assembly **200** has a locked state and a released state, and it is switchable between the locked state and the released state. In the locked state, the snap-fit assembly

200 is engaged with the cover body **2000**; and in the released state, the snap-fit assembly **200** is separated from the cover body **2000**. The first driving assembly **300** is in transmission connection with the pull-cover body **100** and allows the pull-cover body **100** to have a first position and a second position in the vertical direction. The first position is located below the second position. At the first position, the snap-fit assembly **200** is configured to be engaged in the groove **2400** of the cover body **2000**; and at the second position, the cover body **2000** is separated from the can body **1000**. The first driving assembly **300** drives the pull-cover body **100** to move between the first position and the second position.

(71) The first driving assembly **300** may be a motor-driven lifting mechanism or a cylinder-driven lifting mechanism, which is not limited herein. In a specific implementation process, referring to FIG. **15** and FIG. **16**, the first driving assembly **300** drives the pull-cover body **100** to move in the vertical direction, and allows the pull-cover body **100** to have the first position and the second position within its movement stroke. There is a certain distance between the first position and the second position to ensure the cover body **2000** and the pull-cover body **100** to be completely separated from the can body **1000**. It will be appreciated that after the cover body **2000** and the pull-cover body **100** are separated from the can body **1000**, the can body **1000** may be moved to a feeding position of the animal food dispensing device, making it available for pet feeding. The cover body **2000** is defined with a groove, and the snap-fit assembly **200** is engaged with the cover body **2000** through the groove, such that when the first driving assembly **300** drives the pull-cover body **100** to move vertically, it pulls the cover body **2000** to move synchronously. At the same time, the press-can apparatus limits the vertical movement of the can body **1000**. In this way, the pull-cover body **100** can pull the cover body **2000** away from the can body **1000** to open the can.

(72) In the embodiments, as shown in FIG. **14** or FIG. **19**, the can further includes a sealing membrane **3000** for sealing the accommodation chamber. A portion of the sealing membrane **3000** is connected to the cover body **2000**, and at least part of the other portion of the sealing membrane **3000** is connected to the can body **1000**. In this way, when the animal food dispensing device opens the cover body **2000**, the sealing membrane **3000** is separated from the can body **1000** along with the cover body **2000**. Specifically, the press-can apparatus is configured to abut against the top of a support portion of the can body **1000** to limit the vertical movement of the can body **1000**. The pull-cover apparatus cooperates with the cover body **2000** and moves vertically to pull the cover body **2000** away from the can body **1000**. Meanwhile, the sealing membrane **3000** moves synchronously with the cover body **2000** to separate from the can body **1000**, to expose the food inside the accommodation chamber for pets to consume, achieving the effect of automatically opening the food container. In other embodiments, the can body and the cover body can be used directly to seal the food. In the embodiments, the support portion of the can body **1000** is the third flange portion **1210** of the support member **1200**.

(73) In the embodiments, the can is placed in the animal food dispensing device, and it is driven to move under the pull-cover apparatus. The movement of the can may be implemented by a turntable driving the can to rotate, or by manual movement, etc., which is not limited herein. In this case, the pull-cover body **100** is located at a position away from the can, which may be used as an initial position of the pull-cover body **100**. When the animal food dispensing device controls to open the can, the first driving assembly **300** drives the pull-cover body **100** to move downward from the initial position to the first position. This movement is taken as a first stroke of the pull-cover apparatus. It can be understood that, in order for the snap-fit assembly **200** to be smoothly engaged into the groove of the cover body **2000**, the snap-fit assembly **200** remains in the released state until it reaches the first position. When the pull-cover body **100** reaches the first position, the snap-fit assembly **200** is switched from the released state to the locked state and is engaged with the cover body **2000**. After the snap-fit assembly **200** is engaged with the cover body **2000**, the first driving assembly **300** drives the pull-cover body **100** to rise. At the same time, the snap-fit assembly **200** pulls the cover body **2000** to rise synchronously. Furthermore, the press-can

apparatus limits the movement of the can body **1000**, to allow cover body **2000** to separate from the can body **1000**, thereby realizing the opening of the can. The movement of the pull-cover body **100** and the cover body **2000** from the first position to the second position is taken as a second stroke of the pull-cover apparatus. It can be understood that, during the second stroke, when the pull-cover body **100** is at the second position, the snap-fit assembly **200** is in the locked state to prevent the cover body **2000** from falling off. The can body **1000** can be transferred to the feeding position of the animal food dispensing device for the pet to consume.

(74) After the pet has finished eating, the can body **1000** moves from the feeding position to below the pull-cover apparatus. The first driving assembly **300** drives the pull-cover body **100** and the cover body **2000** to descend from the second position to the first position. This movement is taken as a third stroke of the pull-cover apparatus. Understandably, during the third stroke, to prevent the cover body **2000** from falling off, the snap-fit assembly **200** remains in the locked state. After the pull-cover body **100** reaches the first position, the cover body **2000** is engaged with the can body **1000**. The snap-fit assembly **200** is switched from the locked state to the released state, thereby separating from the cover body **2000**. Then, the first driving assembly **300** drives the pull-cover body **100** and the snap assembly **200** to rise to the initial position or the second position. This movement is taken as a fourth stroke of the pull-cover apparatus. It will be appreciated that, in some embodiments, there may be a situation where all the food inside the can body **1000** is consumed; and in this case, the animal food dispensing device may drive the can to a disposal position for discarding. In another embodiment, there may be a situation where there is still food left in the can body **1000**; and in this case, the animal food dispensing device may drive the can to a refrigerated storage position to ensure the quality of the food.

(75) In other embodiments, in the process of the pull-cover body **100** moving to the first position, the snap-fit assembly **200** moves downward with the cover body **2000** to abut against the can body **1000**, and the pull-cover body **100** continues to move downward until it reaches the first position. In this process, the pull-cover body **100** can realize the snap-fit between the cover body **2000** and the can body **1000**.

(76) In some embodiments, referring to FIG. 20 and FIG. 21, the pull-cover apparatus further includes a second driving assembly **400**. The second driving assembly **400** includes a second motor **410** and a first cam **420**. The second motor **410** is connected to the first cam **420**, to drive the first cam **420** to rotate. The outer peripheral wall of the first cam **420** is provided with a protruding portion **421** extending in a direction perpendicular to the axial direction of the second motor **410** and a recessed portion **422**. The snap-fit assembly **200** includes an engaging member **210**, which includes an engaging portion **211** and a pushing portion **212** connected to each other. Specifically, the engaging member **210** extends along the radial direction of the first cam **420**. The pushing portion **212** is arranged at one end of the engaging member **210** close to the first cam **420** for abutting against the protruding portion **421** and the recessed portion **422**, and the engaging portion **211** is arranged at the other end of the engaging member **210** away from the first cam **420** for engagement with the cover body **2000**. In the locked state, the first cam **420** drives the protruding portion **421** to abut against the pushing portion **212**, and the pushing portion **212** drives the engaging portion **211** to be engaged in the groove **2400** of the cover body **2000**. In the released state, the first cam **420** drives the recessed portion **422** to abut against the pushing portion **212**, and the pushing portion **212** drives the engaging portion **211** to be separated from the cover body **2000**.

(77) In a specific implementation process, with reference to FIG. 17 and FIG. 22, an output shaft of the second motor **410** is connected to the first cam **420**, and drives the first cam **420** to rotate. During the rotation of the first cam **420**, the protruding portion **421** and the recessed portion **422** of the first cam **420** alternately abut against the pushing portion **212** of the engaging member **210**, such that the snap-fit assembly **200** is switched between the locked state and the released state. In the embodiments, multiple engaging members **210** are arranged at intervals along the circumferential direction of the first cam **420** to ensure that the cover body **2000** is smoothly pulled

after the engaging members **210** are engaged with the cover body **2000**. Correspondingly, the numbers of the protruding portions **421** and the recessed portions **422** are the same as or more than the number of the engaging members **210** to ensure that each engaging member **210** abuts against one protruding portion **421** or one recessed portion **422** during the rotation of the cam. In the embodiments, the engaging member **210** is engaged in the groove of the cover body **2000**. In the embodiments, as shown in FIG. **16**, in the locked state, the snap-fit assembly **200** is engaged in the groove **2400**; and in the released state, the snap-fit assembly **200** is separated from the groove **2400**. It can be understood that the number of the grooves defined in the cover body **2000** is the same as or greater than the number of the engaging members **210**, or the groove of the cover body **2000** is an integral one, ensuring that when in the locked state, each engaging member **210** is engaged within the corresponding groove.

(78) In a specific implementation process, the axis of the second cam **430** is coincident with the axis of the pull-cover body **100**. The extension direction of the engaging member **210** is consistent with the radial direction of the second cam **430**, and the engaging member **210** is movable along the radial direction. Further, the pull-cover body **100** is in the shape of a sleeve having an internal first accommodation space, and the second driving assembly **400** and the snap-fit assembly **200** are arranged in the first accommodation space. This allows for the efficient use of space and helps to reduce the overall volume of the animal food dispensing device. A lower wall surface of the first accommodation space is provided with a guide rail, and the engaging member **210** is slidably fitted with the guide rail. Specifically, in some embodiments, the lower wall surface of the pull-cover body **100** is opened with a sliding groove, the engaging member **210** is slidably connected in the sliding groove. The inner wall of the sliding groove forms the guide rail, to allow the engaging member **210** to slide along the inner wall of the sliding groove. Specifically, in some embodiments, the side of the lower wall surface of the pull-cover body **100** facing away from the first accommodation space is connected with a limiting plate **120**, the bottom of the pull-cover body **100** is recessed upward to form the first accommodation space, and the limiting plate **120** is arranged at the bottom of the first accommodation space. In this case, the upper surface of the limiting plate **120** forms the lower wall surface of the first accommodation space. In another embodiment, the side of the lower wall surface of the pull-cover body **100** facing away from the first accommodation space is connected with a limiting plate **120**, the bottom of the pull-cover body **100** is recessed toward the direction of the first accommodation space, to form the sliding groove between the limiting plate **120** and the pull-cover body **100**. The engaging member **210** slides in the sliding groove, the inner wall of the sliding groove and the limiting plate **120** form the guide rail, and the engaging member **210** moves along the sliding groove.

(79) As shown in FIG. **23**, a second avoiding groove **121** is defined in the center of the limiting plate **120**. The second avoiding groove **121** is configured to avoid the handle portion **2500** of the cover body **2000** as shown in FIG. **16**.

(80) In the embodiments, the first cam **420** rotates to cause the protruding portion **421** to abut against the pushing portion **212**, which in turns drives the engaging member **210** to move, and allows the engaging portion **211** to extend out of the pull-cover body **100** to engage with the cover body **2000**. As the first cam **420** continues to rotate to allow the recessed portion **422** to face the pushing portion **212**, the engaging member **210** needs to retract to allow the pushing part **212** to abut against the recessed portion **422**. In the embodiments, the snap-fit assembly **200** further includes a reset member **220**. Specifically, the reset member **220** has a certain elasticity, and the reset member **220** is connected to the engaging member **210**. When the protruding portion **421** drives the engaging member **210** to move, the reset member **220** is compressed and thus has a certain elastic force. As the first cam **420** continues to rotate to allow the recessed portion **422** to face the pushing portion **212**, the reset member **220** returns to its original state and releases the elastic force, thereby driving the engaging member **210** to move in the opposite direction. This action cooperates with the rotation of the first cam **420** to allow the pushing portion **212** to abut

against the recessed portion **422**. The reset member **220** may be an elastic material, such as rubber, silicone, or a spring, which is not limited in the embodiments.

(81) In a specific implementation process, the engaging member **210** is further provided with a fixing portion **215** extending upward along the axial direction of the second motor **410**. One end of the reset member **220** is connected with the pull-cover body **100**, and the other end of the reset member **220** is connected with the fixing portion **215**. The reset member **220** applies an elastic resetting force to the engaging member **210**, allowing it to switch from the locked state to the released state. Specifically, the inner wall of the first accommodation space of the pull-cover body **100** is provided with a mounting plate. The lower wall surface of the first accommodation space is defined with an elongated hole, and the extension direction of the elongated hole is consistent with the extension direction of the engaging member **210**. The top surface of the engaging member **210** protrudes upward to form the fixing portion **215**, and the fixing portion **215** is slidable in the elongated hole. One end of the reset member **220** is connected with the mounting plate, and the other end of the reset member **220** is connected with the fixing portion **215**, such that the engaging member **210** compresses the reset member **220** during its movement from the released state to the locked state, thereby allowing the reset member **220** to have a restoring force to return to its original state. When the engaging member **210** is switched from the locked state to the released state, the protruding portion **421** of the first cam **420** gradually moves away from the pushing portion **212**. In this case, the reset member **220**, due to its restoring force, returns to its original state, thereby driving the engaging member **210** to move in the opposite direction. In this way, the pushing portion **212** of the engaging member **210** abuts against the recessed portion **422** of the first cam **420**; that is, the engaging member **210** is in the released state. It can be understood that the end surface of the pushing portion **212** and the side surface of the protruding portion **421** are both arc-shaped, to ensure a smooth switch between the locked state and the released state of the engaging member **210**.

(82) In the embodiments, as shown in FIG. 23, the engaging portion **211** is provided with a first guide inclined surface **213** and a support surface **214** connected to each other. The first guide inclined surface **213** extends upward from the edge of the engaging member **210** to the support surface **214**. When the pull-cover body **100** moves toward the first position, the first guide inclined surface **213** is configured to cooperate with the cover body **2000** to guide the cover body **2000** into the support surface **214**.

(83) It can be understood that, in practice, the placement position of the can is not precise, yet the pull-cover body **100** needs to cooperate with the cover body **2000** of the can to ensure that the engaging portion **211** of the engaging member **210** can be smoothly engaged in the groove of the cover body **2000**. In the embodiments, the first guide inclined surface **213** is cooperated with the cover body **2000**, which is conducive to the cooperation between the peripheral surface of the pull-cover body **100** and the top of the cover body **2000**, making it easier for the engaging portion **211** to insert into the cover body **2000**. Additionally, the support surface **214** abuts against the cover body **2000**, to ensure that the pull-cover body **100** can smoothly pull the cover body **2000**.

(84) Further, the bottom end of the pull-cover body **100** is provided with a guide portion **110** for cooperating with the cover body **2000**. The guide portion **110** is configured to move in the inner peripheral wall of the cover body **2000**, to allow the pull-cover body **100** to move into or out of the cover body **2000**. The guide portion **110** is defined with a first through hole **111**. In the locked state, the engaging portion **211** can extend out of the first through hole **111** and engage with the groove **2400** of the cover body **2000**. In a specific implementation process, the guide portion **110** is arranged at a bottom edge of the pull-cover body **100**, and is arranged along the circumferential direction of the pull-cover body **100**. In the released state, the engaging portion **211** is located in the first through hole **111**, or, an end of the engaging portion **211** is flush with the guide portion **110**, preventing the engaging portion **211** from affecting the entry of the pull-cover body **100** into the cover body **2000**.

(85) In a more specific implementation process, as shown in FIG. 4, the cover body **2000** includes a side wall portion **2100**, a cover tray portion **2200**, and a cover flange portion **2300**. The cover tray portion **2200** is connected to the inner peripheral wall of the side wall portion **2100**. The cover tray portion **2200** allows the cover body **2000**, along an up-down direction of the cover body **2000**, to be defined with a recessed space having an opening opened upward and a lower space having an opening opened downward. The cover flange portion **2300** is arranged in the recessed space. One end of the cover flange portion **2300** is connected with the side wall portion **2100**, and the other end of the cover flange portion **2300** extends inward in a direction away from the cover body **2000**. The cover flange portion **2300**, the side wall portion **2100** and the cover tray portion **2200** together form the groove **2400**. In the locked state, the guide portion **110** extends into the groove **2400** and is engaged with the cover flange portion **2300**, such that when the pull-cover body **100** rises, it can pull the cover body **2000** to move.

(86) Further, the guide portion **110** is provided with a second guide inclined surface **112**. The second guide inclined surface **112** is inclined from the bottom of the pull-cover body **100** in a vertical upward direction toward the direction close to the press-can apparatus. When the pull-cover body **100** moves to the first position, the second guide inclined surface **112** is configured to cooperate with the cover body **2000**, to move the pull-cover body **100** into the cover body **2000**. Specifically, the second guide inclined surface **112** and the first guide inclined surface **213** are both located at the edge of the cover body **2000**; besides, the second guide inclined surface **112** is connected to the outer side wall and the outer bottom wall of the pull-cover body **100** and is in contact with the cover body **2000**, serving as a guide. This allows the pull-cover body **100** to smoothly enter and exit the cover body **2000**. In the embodiments, the second guide inclined surface **112** is cooperatively arranged with the first guide inclined surface **213**, and the two have the consistent inclination direction and inclination angle, to allow the bottom of the pull-cover body **100** to smoothly enter and exit the cover body **2000**.

(87) In some embodiments, the second driving assembly **400** further includes a second cam **430**. The first cam **420** includes a cam shaft **423**, and the second cam **430** is sleeved on the cam shaft **423**. The second motor **410** drives the first cam **420** and the second cam **430** to move synchronously. The animal food dispensing device further includes a first detection assembly **130**, which includes a first signal emitter **131** and a first signal receiver **132**. The first signal emitter **131** is fixedly connected to the second cam **430**, and the first signal receiver **132** is fixedly connected to the pull-cover body **100**. In the locked state or/and the released state, the first signal emitter **131** is provided in correspondence with the first signal receiver **132**.

(88) In a specific implementation process, the second cam **430** is located above the first cam **420**, and the cam shaft **423** of the first cam **420** extends upward along the axis direction, to allow the second cam **430** to be sleeved on the camshaft **423**. Besides, a limiting surface is provided between the cam shaft **423** and the second cam **430**, to prevent circumferential rotational displacement between the first cam **420** and the second cam **430** during rotation, thereby allowing the second motor **410** to drive the first cam **420** and the second cam **430** to rotate synchronously. In the embodiments, the first detection assembly **130** is configured to detect the action of the snap-fit assembly **200**. Specifically, through the rotation of the second cam **430**, the first signal emitter **131** triggers the first signal receiver **132**, achieving detection on the action of the snap-fit assembly **200**. Specifically, the first signal emitter **131** is fixedly connected with the second cam **430**, and the second cam **430** moves synchronously with the first cam **420**. When the first cam **420** drives the protruding portion **421** to allow the engaging member **210** in the locked state, the first signal emitter **131** fixedly connected with the second cam **430** is provided in correspondence with the first signal receiver **132**; and in this case, the first detection assembly detects that the snap-fit assembly **200** is in the locked state. Or, when the first cam **420** drives the recessed portion **422** to allow the engaging member **210** in the released state, the first signal emitter **131** fixedly connected with the second cam **430** is provided in correspondence with the first signal receiver **132**; and in this case,

the first detection assembly detects that the snap-fit assembly **200** is in the released state.

(89) In the embodiments, referring to FIG. **17**, the first detection assembly **130** includes two first signal emitters **131**. The first signal emitter **131** located above the protruding portion **421** is configured to detect whether the snap-fit assembly is in the locked state, and the first signal emitter **131** located above the recessed portion **422** is configured to detect whether the snap-fit assembly is in the released state.

(90) In the embodiments, referring to FIG. **17**, the first detection assembly **130** is mechanical switches. The first signal emitter **131** is a mechanical protrusion, and the first signal receiver **132** is a mechanical button. Specifically, the mechanical protrusion is arranged on the peripheral side wall of the second cam **430**. In practice, the protrusion is arranged on the peripheral side wall of the second cam **430**; and the mechanical button is arranged on the pull-cover body **100** and located on one side of the second cam **430**. In this way, the rotation of the second cam **430** causes the mechanical protrusion to contact the mechanical button, which in turn triggers the first signal receiver **132** to generate a first detection signal. As needed, multiple mechanical protrusions may be spaced apart on the peripheral side wall of the second cam **430** along the circumferential direction. In this way, one full rotation of the second cam **430** can cause multiple contacts between the mechanical protrusions and the mechanical button, meeting the detection requirements for the snap-fit assembly **200**. In the embodiments, the number of the mechanical protrusions is the same as the number of the protruding portions **421**, and the angle of the extension protruding direction of the mechanical protrusions is consistent with that of the protruding portions **421**.

(91) Further, the first detection assembly **130** includes two first signal receivers **132**. The number of the first signal emitters **131** is the same as the number of the engaging members **210** and is denoted as N . The first signal emitters **131** and the engaging members **210** are evenly spaced along the axis of the second motor **410** within a projection on a horizontal plane. The line connecting the axis of the second motor **410** to one of the first signal receivers **132** is $L1$, and the line connecting the axis of the second motor **410** to the other first signal receiver **132** is $L2$, wherein the central angle defined between $L1$ and $L2$ is α , and α and N satisfy the following equation: $\alpha=180/N$.

(92) In the embodiments, N is 4 as an example, then, α is 45° , and the first signal emitters **131** are evenly spaced on the second cam **430**. When the first signal emitter **131** triggers the first signal receiver **132** located above the protruding portion **421**, the first cam **420** drives the engaging member **210** to extend to engage with the groove **2400** of the cover body **2000**. In this case, the snap-fit assembly **200** is in the locked state, and the first driving assembly **300** can drive the pull-cover body **100** to rise and drive the cover body **2000** to rise synchronously. The second cam **430** continues to rotate by 45 degrees. The first signal emitter **131** triggers the first signal receiver **132** located above the recessed portion **422**, and the engaging member **210** cooperates with the recessed portion **422** of the first cam **420**. In this case, the snap-fit assembly **200** is in the released state, and the first driving assembly **300** can further drive the pull-cover body **100** away from the cover body **2000**. During this process, the pull-cover body **100** moves from the first position to the second position to pull the cover body **2000** away from the can body **1000**; then, it moves back from the second position to the first position to place the cover body **2000** back onto the can body **1000**.

(93) Further, the first driving assembly **300** includes a first motor **310** and a first transmission member which are in transmission connection with the pull-cover body **100**. The first motor **310** drives the first transmission member to move vertically. The pressing member **600** is in the shape of a sleeve with an internal second accommodation space. One end of the pressing member **600** close to the can is open, and the other of the pressing member **600** is defined with a second through hole penetrating up and down. The pull-cover body **100** is arranged in the second accommodation space. The first transmission member passes through the second through hole **611** to drive the pull-cover body **100** to move vertically.

(94) The pull-cover body **100** is located in the second accommodation space of the pressing member **600** and moves in the second accommodation space. It can be understood that the bottom

end of the pressing member **600** is open, which facilitates the connection between the pull-cover body **100** and the cover body **2000** to open the cover body **2000**. The pressing member **600** is sleeved outside the pull-cover body **100**, and the bottom surface of the pressing member **600** abuts against the support portion of the can body **1000**. It can be understood that, the outer wall of the support portion is located outside the pressing member **600** within a projection on a horizontal plane, to ensure effective abutment between the pressing member **600** and the support portion, thereby effectively limiting the vertical movement of the can body **1000**.

(95) As shown in FIG. **21**, the top of the pressing member **600** is opened with the second through hole **611**. In some embodiments, the pressing member **600** is provided with an integrally formed top, the second through hole **611** is arranged on the top. The first motor **310** is mounted on the top of the pressing member **600** and located on the surface facing away from the second accommodation space. In another embodiment, the animal food dispensing device further includes a fixing member **610**. The pressing member **600** and the pull-cover body **100** are fixedly connected to the fixing member **610**. The fixing member **610** is connected to the top of the pressing member **600** by means of screw connection or engagement connection, such that the second accommodation space is closed at its top. The first motor **310** is mounted on the side of the fixing member **610** facing away from the fixing member **610**. The first motor **310** drives the pull-cover body **100** to move between the first position and the second position by the first transmission member. The first transmission member may be a gear transmission mechanism, or a conveyor belt mechanism, or the like, which is not limited herein, as long as it enables the pull-cover body **100** to move up and down under the driving action of the first motor **310**. In the embodiments, the fixing member **610** is connected to both the pressing member **600** and the pull-cover body **100**, such that the movement of the pressing member **600** drives the pull-cover body **100** to move. Specifically, according to the animal food dispensing device in the embodiments, in the action of separating the can body **1000** from the cover body **2000**, the pressing member **600** moves downward to abut against the support portion of the can body **1000**. The pressing member **600** drives the pull-cover body **100** to move downward, and then the pull-cover body **100** moves downward to reach the first position. After the snap-fit assembly **200** is engaged with the cover body, the pull-cover body **100** moves upward to achieve the separation of the can body **1000** and the cover body **2000**.

(96) In the embodiments, the bottom of the pressing member **600** is provided with a flanging **620** extending outward along the circumferential direction. The flanging **620** is configured to abut against the support portion of the can body **1000**, to increase the contact area between the pressing member **600** and the support portion. In the embodiments, to ensure the effectiveness of the abutment, the top surface of the support portion is flat, which is convenient for the flanging **620** to abut against the support portion. Specifically, the bottom end of the pressing member **600** bends away from the axis to form the flanging **620**. The pressing member **600** surrounds the pull-cover body **100** and the cover body **2000** inside the second accommodation space. The flanging **620** abuts against the support portion to limit the upward vertical displacement of the can body **1000**.

(97) Further, the fixing member **610** is opened with a second through hole **611**. The first transmission member includes a first gear **320** and a first rack **330**. The first motor **310** is mounted on the fixing member **610**. An output shaft of the first motor **310** is in transmission connection with the first gear **320**. The first rack **330** has a first surface and a second surface facing the first surface. The first surface is meshed with the first gear **320** for transmission. The surface of the fixing member **610** facing the first motor **310** is provided with a limiting portion **612**, and the limiting portion **612** is arranged close to the second surface to limit the first rack **330** to move in a direction away from the first gear **320**. The first rack **330** passes through the second through hole **611** to drive the pull-cover body **100** to move vertically.

(98) In a specific implementation process, the first rack **330** extends in the vertical direction. One end of the first rack **330** is connected to the top of the pull-cover body **100**, and the other end of the first rack **330** passes through the second through hole **611**, and the first rack **330** is allowed to slide

along the second through hole **611**. The first motor **310** drives the first rack **330** to move in the vertical direction by the first gear **320**, to allow the pull-cover body **100** to move between the first position and the second position. The fixing member **610** is connected with a limiting portion **612**, which is in contact with the second surface of the first rack **330**, to prevent horizontal displacement of the first rack **330**. On the one hand, this can prevent the pull-cover body **100** from swaying in the horizontal direction, enhancing the movement stability of the first rack **330** and the pull-cover body **100**. On the other hand, it allows the first rack **330** to be always meshed with the first gear **320**, ensuring the effectiveness of the first motor **310** driving the movement of the first rack **330**. It can be understood that the first surface of the first rack **330** is provided with gear teeth for meshing with the first gear **320**, and the second surface of the first rack **330** is flat to facilitate cooperation with the limiting portion **612**.

(99) In some embodiments, the limiting portion **612** includes a third roller **613**. The third roller **613** is in rolling operation with the first rack **330** to reduce friction. Specifically, the peripheral side wall of the third roller **613** is in contact with the second surface of the first rack **330**, such that the first rack **330** is in rolling operation with the third roller **613**, thereby reducing friction. In a specific implementation process, the third roller **613** is fixed on the fixing member **610** and rotatable relative to the fixing member **610**. In another embodiment, the limiting portion **612** includes a rolling ball that is in contact with the first rack **330**, which can further reduce friction.

(100) In some embodiments, the animal food dispensing device further includes a second detection assembly **630**, which includes a second signal emitter **631** and a second signal receiver **632**. The second signal emitter **631** is fixedly connected to the first rack **330**, and the second signal receiver **632** is fixedly connected to the fixing member **610**. The second detection assembly **630** is configured to cooperate with the first driving assembly **300**, to control the movement of the pull-cover body **100**.

(101) The second detection assembly **630** may be a photoelectric sensor, a mechanical switch, or the like, which is not limited herein, as long as it ensures smooth triggering. In a specific implementation process, the second detection assembly **630** is a mechanical switch, such as a limit switch. Specifically, the second signal emitter **631** is a mechanical protrusion, and the second signal receiver **632** is a mechanical button. At the first position, the second signal emitter **631** abuts against the second signal receiver **632**. When the first rack **330** moves to a position where the second signal emitter **631** triggers the second signal receiver **632**, the second signal receiver **632** is triggered to generate a second detection signal. In the embodiments, the second detection signal is the signal indicating that the pull-cover body **100** has reached the first position.

(102) In some embodiments, there are two second signal emitters **631** and two second signal receivers **632**. One second signal emitter **631** is arranged on the first rack **330** and located in the space above the fixing member **610**, and one second signal receiver **632** is mounted on the fixing member **610** and located on the side of the fixing member **610** facing away from the second accommodation space; and in this case, the second signal emitter **631** is configured to trigger the second signal receiver **632**. The other second signal emitter **631** is arranged on the first rack **330** and located in the space below the fixing member **610**, or the other second signal emitter **631** is arranged on the top surface of the pull-cover body **100**, and the other second signal receiver **632** is arranged on the fixing member **610** and located on the side of the fixing member **610** facing the second accommodation space; and in this case, the second signal emitter **631** is configured to trigger the second signal receiver **632**. In another embodiment, one second signal emitter **631** is mounted on the first rack **330**, and two second signal receivers **632** with a vertical spacing distance therebetween are arranged on the fixing member **610**. When the pull-cover body **100** is at the first position, the second signal emitter **631** triggers one second signal receiver **632**; and when the pull-cover body **100** is at the second position, the second signal emitter **631** triggers the other second signal receiver **632**.

(103) Based on the aforementioned, when the pull-cover body **100** moves to the first position, the

second signal emitter **631** is provided in correspondence with the second signal receiver **632** to generate a detection signal. When the pull-cover body **100** moves to the second position, the other second signal emitter **631** is provided in correspondence with the other second signal receiver **632** to generate a detection signal.

(104) In some embodiments, the press-can apparatus includes a pressing member **600** and a third driving assembly **500**, and an outer peripheral wall of the can body **1000** is provided with a support portion. The third driving assembly **500** is in transmission connection with the pressing member **600**, to allow the pressing member **600** to have a third position and a fourth position in the vertical direction. At the third position, the pressing member **600** abuts against the support portion to limit vertical movement of the can body **1000**. At the fourth position, the pressing member **600** is separated from the support portion. The third driving assembly **500** drives the pressing member **600** to move between the third position and the fourth position.

(105) In a specific implementation process, the support portion is fixed on the outer peripheral wall of the can body **1000** and protrudes from the outer peripheral wall, facilitating the pressing member **600** to move from top to bottom and abuts against the top surface of the support portion. Specifically, the bottom of the pressing member **600** abuts against the top surface of the support portion. The third driving assembly **500** drives the pressing member **600** to move up and down. The third position is located below the fourth position. The third driving assembly **500** drives the pressing member **600** to move from the fourth position to the third position, to allow the pressing member **600** to move from the direction away from the support portion toward the direction close to the support portion, thereby abutting against the support portion. This process serves as a fifth stroke of the pressing member **600**. The third driving assembly **500** drives the pressing member **600** to move from the third position to the fourth position, to allow the pressing member **600** to move from the support portion in a direction away from the support portion. This process serves as a sixth stroke of the pressing member **600**. The third driving assembly may be a motor-driven structure, or may be a cylinder lifting mechanism, which is not limited herein.

(106) Further, the third driving assembly **500** includes a third motor **510** and a second transmission member. The second transmission member includes a second gear **520** and a second rack **530**. An output shaft of the third motor **510** is in transmission connection with the second gear **520**, the pressing member **600** is fixedly connected with the second rack **530**, and the second gear **520** is meshed with the second rack **530** to drive the pressing member **600** to move in the vertical direction.

(107) In the embodiments, the third motor **510** drives the pressing member **600** to move between the third position and the fourth position by the second transmission member. Specifically, the second rack **530** extends vertically and is fixedly installed on the outer side wall of the pressing member **600**. The third motor **510** drives the second gear **520** to rotate, such that the second rack **530** moves in the vertical direction, thereby realizing the movement of the pressing member **600** between the third position and the fourth position.

(108) In some embodiments, the animal food dispensing device further includes a housing **900** in the shape of a sleeve with an internal third accommodation space. The third motor **510** is mounted on the housing **900**, and the press-can apparatus and the pull-cover apparatus are arranged in the third accommodation space. In a specific implementation process, the housing **900** is sleeved outside the pressing member **600** accommodated in the third accommodation space, and the third motor **510** is mounted on the housing **900**. The housing **900** is defined with a first avoiding groove **910** to avoid the second rack **530**, allowing the second rack **530** to slide in the first avoiding groove **910**.

(109) It can be understood that the pull-cover body **100** is located in the second accommodation space of the pressing member **600**, and the pull-cover body **100** moves up and down in the second accommodation space. In this way, there is a certain gap between the outer wall of the pull-cover body **100** and the inner wall of the second accommodation space, which may cause the pull-cover

body **100** to sway during its vertical movement and fail to enter the cover body **2000**. This could result in that the engaging member **210** is unable to engage with the cover body **2000**, thereby failing to open the can. In the embodiments, the animal food dispensing device further includes a first walking assembly **700**. The first walking assembly **700** includes a first roller **710**, which is assembled to the pull-cover body **100**. The first roller **710** may be in rolling cooperation with the pressing member **600** to reduce friction generated when the pull-cover body **100** and the pressing member **600** move relative to each other.

(110) Referring to FIG. **21** and FIG. **23**, in the embodiments, the first walking assembly **700** is arranged on the pull-cover body **100** and contacts the pressing member **600** to limit the horizontal displacement of the pull-cover body **100**, thereby providing a good limiting and guiding effect on the pull-cover body **100**. Specifically, the first walking assembly **700** further includes a first mounting base **720**, which is fixed on the side wall of the pull-cover body **100** by a screw. The first roller **710** is mounted in the first mounting base **720**, and the outer peripheral wall of the first roller **710** is in contact with the inner wall of the pressing member **600**. During the process of the pull-cover body **100** moving up and down, the first roller **710** rolls along the pressing member **600**, which reduces the friction between the pull-cover body **100** and the pressing member **600**, and also limits the pull-cover body **100**, thereby enhancing the stability of the up and down movement of the pull-cover body **100**. It can be understood that, according to actual needs, the first mounting base **720** may be provided with multiple first rollers **710** to further improve the stability of the pull-cover body **100**. In addition, multiple first walking assemblies **700** are spaced apart along the circumferential direction of the pull-cover body **100** to ensure the balance of the pull-cover body **100**. The first roller **710** may be made of a soft material such as rubber or silicone, which reduces wear and noise, and also helps to save costs.

(111) The animal food dispensing device further includes a second walking assembly **800**. The second walking assembly **800** includes a second roller **810**, which is assembled on the housing **900**. The second roller **810** may be in rolling operation with the pressing member **600** to reduce the friction generated when the pressing member **600** moves relative to the housing **900**.

(112) Referring to the aforementioned, in a specific implementation process, the pressing member **600** moves in the third accommodation space of the housing **900**. In this way, there is also a certain gap between the outer wall of the pressing member **600** and the inner wall of the third accommodation space, which may cause the pressing member **600** to sway during its vertical movement and fail to accurately abut against the support portion of the can body **1000**. This could result in that the pressing member **600** is unable to limit the can body **1000**, thereby failing to open the can. In the embodiments, the second walking assembly **800** is arranged between the pressing member **600** and the housing **900** to limit the horizontal displacement of the pressing member **600**, ensuring smooth opening of the can body **1000**.

(113) Specifically, the second walking assembly **800** further includes a second mounting base **820**, which is fixed on the side wall of the pressing member **600** by a screw. The second roller **810** is mounted in the second mounting base **820**, and the outer peripheral wall of the second roller **810** is in contact with the inner wall of the housing **900**. During the process of the pressing member **600** moving up and down, the second roller **810** rolls along the pressing member **600**, which reduces the friction between the pressing member **600** and the housing **900**, and also limits the pressing member **600**, thereby enhancing the stability of the up and down movement of the pressing member **600**. It can be understood that, according to actual needs, the second mounting base **820** may be provided with multiple second rollers **810** to further improve the stability of the pressing member **600**. In addition, multiple second walking assemblies **800** are spaced apart along the circumferential direction of the pressing member **600** to ensure the balance of the pressing member **600**. The second roller **810** may be made of a soft material such as rubber or silicone, which reduces wear and noise, and also helps to save costs.

(114) In some embodiments, the animal food dispensing device further includes a third detection

assembly **920**. The third detection assembly **920** includes a third signal emitter **921** and two third signal receivers **922**. The third signal emitter **921** is fixedly connected to the press-can apparatus, and the third signal receivers **922** are fixedly connected to the housing **900**. At the third position, one of the two third signal receivers **922** is provided in correspondence with the third signal receiver **922**; and at the fourth position, the other of the two third signal receivers **922** is provided in correspondence with the third signal receiver **922**.

(115) The third detection assembly **920** cooperates with the third driving assembly **500** to control the movement of the pressing member **600**. Specifically, the third signal emitter **921** may be arranged on the second rack **530**, and moves with the second rack **530**; or, the second rack **530** serves as the third signal emitter **921**. The two third signal receivers **922** are arranged on the housing **900**, and there is a certain distance, namely the movement distance of the pressing member **600**, between the two third signal receivers **922**. In the embodiments, the pressing member **600** reaches the fourth position, the rack is located between the two third signal receivers **922**, and the pressing member **600** moves upward, so that when the top end of the rack **922** is sensed by one third signal receiver, this third signal receiver **922** is triggered to generate a third detection signal to determine the position condition of the pressing member **600**. Similarly, the pressing member **600** is located at the third position, when the bottom end of the rack is sensed by the other third signal receiver **922**, this third signal receiver **922** is triggered, the third motor **510** stops, and a third detection signal is generated to determine the position condition of the pressing member **600**.

(116) In a specific implementation process, the third detection assembly **920** is a photoelectric sensor, the third signal emitter **921** is a photoelectric emitter, and the third signal receiver **922** is a photoelectric receiver; or, the third detection assembly **920** is a magnetic field sensor, the third signal emitter **921** is a magnetic signal emitter, and the third signal receiver **922** is a magnetic signal receiver; or, the third detection assembly **920** is a mechanical switch, the third signal emitter **921** is a mechanical protrusion, and the third signal receiver **922** is a mechanical button, which is not detailed herein.

(117) The foregoing are only optional embodiments of the present disclosure, and are not intended to limit the patent scope of the present disclosure. Under the inventive concept of the present disclosure, any equivalent structural transformation made by using the contents of the specification and drawings of the present disclosure, or directly/indirectly applied in other related technical fields, are included in the patent protection scope of the present disclosure.

Claims

1. A food container, comprising: a can body comprising a container body, the container body comprising a container bowl portion, the container bowl portion being defined with an accommodation chamber for holding food and an opening opened upward and communicated with the accommodation chamber; and a cover body, the cover body comprising a side wall portion, a cover tray portion, and a cover flange portion; the cover tray portion being connected to an inner peripheral wall of the side wall portion, the cover tray portion allowing the cover body, along an up-down direction of the cover body, to be defined with a recessed space having an opening opened upward and a lower space having an opening opened downward, the cover flange portion being arranged in the recessed space, an end of the cover flange portion being connected to the side wall portion, the other end of the cover flange portion extending inward in a direction away from the side wall portion; and the cover flange portion, the side wall portion, and the cover tray portion defining a groove; wherein the cover body is assembled to the can body, and the container body is snap-fitted with the side wall portion; wherein the food container further comprises: a sealing membrane for sealing the accommodation chamber; wherein a portion of the sealing membrane is connected to the cover tray portion, and at least part of the other portion of the sealing membrane is connected to the container bowl portion.

2. The food container according to claim 1, wherein the sealing membrane comprises: a sealing membrane body; and a connection lug portion extending outward from an outer peripheral edge of the sealing membrane body; wherein the sealing membrane body is connected to an upper surface of the container bowl portion, and the connection lug portion is folded upward and inward to be connected to a lower surface of the cover tray portion.
 3. The food container according to claim 2, wherein the connection lug portion comprises a first adhesion section and a second adhesion section connected to the first adhesion section, and the first adhesion section is closer to an outer peripheral edge of the sealing membrane body than the second adhesion section; and the connection lug portion is one of two connection lug portions, the first adhesion section of one of the two connection lug portions is connected to the lower surface of the cover tray portion, and the second adhesion section of the other one of the two connection lug portions is connected to the lower surface of the cover tray portion.
 4. The food container according to claim 2, wherein the connection lug portion is one of a plurality of connection lug portions, and the plurality of connection lug portions are symmetrically arranged about a central axis of the sealing membrane.
 5. The food container according to claim 1, wherein the sealing membrane has a first surface and a second surface facing the first surface, the first surface of the sealing membrane is connected to an upper surface of the container bowl portion, and the second surface of the sealing membrane is connected to a lower surface of the cover tray portion; or, the cover tray portion is defined with a first through hole, the sealing membrane comprises a sealing membrane body and a connection lug portion extending outward from an outer peripheral edge of the sealing membrane body, the sealing membrane body is connected to an upper surface of the container bowl portion, the connection lug portion passes through the first through hole into the recessed space, and the connection lug portion is connected to an upper surface of the cover tray portion.
 6. The food container according to claim 1, wherein the cover body is further provided with a handle portion arranged in the recessed space, the handle portion is connected to the cover tray portion, and an upper surface of the handle portion is lower than the cover flange portion in an axial direction of a central axis of the cover body.
 7. The food container according to claim 1, wherein the cover body is further provided with a first flange portion arranged in the lower space, an end of the first flange portion is connected to the side wall portion, and the other end of the first flange portion extends inward in a direction away from the side wall portion; and the container body is further provided with a second flange portion, an end of the second flange portion is connected to the container bowl portion, and the other end of the second flange portion extends outward in a direction away from the container body; wherein the cover body is assembled to the can body, and the first flange portion is snap-fitted with the second flange portion.
 8. The food container according to claim 7, wherein the second flange portion forms an upper peripheral edge of the container body; and the cover body is assembled to the can body, and an inner edge of the cover flange portion fails to protrude beyond an inner surface of the second flange portion within a projection of the food container on a horizontal plane.
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